

Machine Learning

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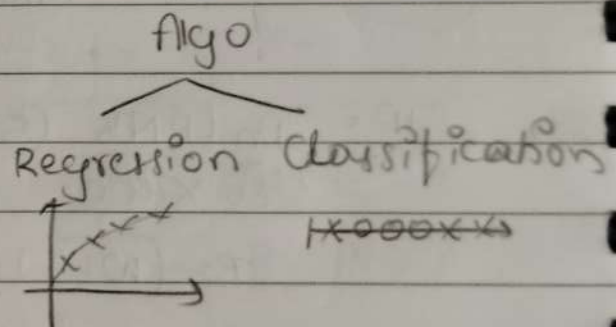
It is the study of computer to learn by itself without being explicitly programmed.

- Arthur (1959)

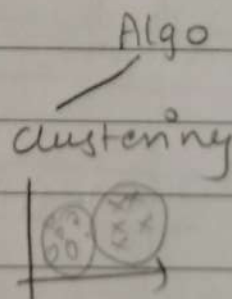
- Supervised Learning
- Unsupervised Learning
- Recommended System
- Reinforcement Learning

Supervised Learning — Model learns from right answer

$X \implies Y$
Input Output

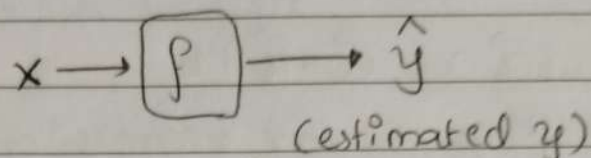


Unsupervised Learning — Find something interesting in unlabeled data.



Linear RegressionTerminology

Training Set



imp
 $f_{w,b}(x) = wx + b$ — y-intercept
 Linear regression with one variable
 or
 Univariate Linear Regression

 w, b : ParametersCost function

$$\text{imp } J(w, b) = \frac{1}{2m} \sum_{i=0}^m (\hat{y}^{(i)} - y^{(i)})^2$$

Goal $\rightarrow$ minimize $J(w, b)$
 w, b

Gradient Descent

Learning Rate.

$$\text{imp } w = w - \alpha \frac{\partial}{\partial w} J(w, b)$$

$$b = b - \alpha \frac{\partial}{\partial b} J(w, b)$$

$$\frac{\partial}{\partial w} J(w, b) = \frac{1}{m} \sum_{i=0}^m (f(x^{(i)}) - y^{(i)}) \cdot x^{(i)}$$

$$\frac{\partial}{\partial b} J(w, b) = \frac{1}{m} \sum_{i=0}^m (f(x^{(i)}) - y^{(i)})$$

For n -number of features

$$\vec{w} = [w_1, w_2, w_3, \dots, w_n]$$

$$\vec{x} = [x_1, x_2, x_3, \dots, x_n]$$

$$f_{\vec{w}, b}(\vec{x}) = \vec{w} \cdot \vec{x} + b$$

multiple linear regression

Feature Scaling

Mean Normalisation

Z-score Normalisation

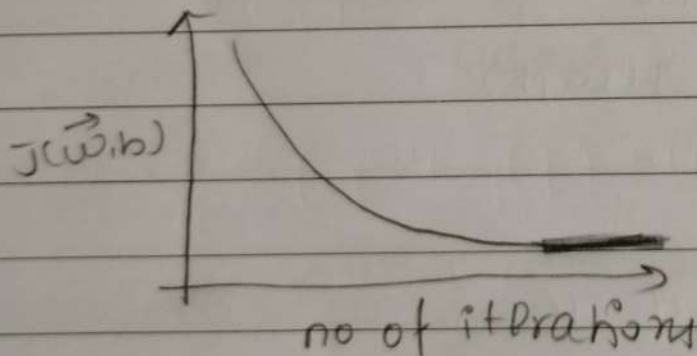
$$x_1 = \frac{x_1 - \mu}{\text{max} - \text{min}}$$

mean

$$x_1 = \frac{x_1 - \mu}{\sigma}$$

Standard deviation

Learning Curve



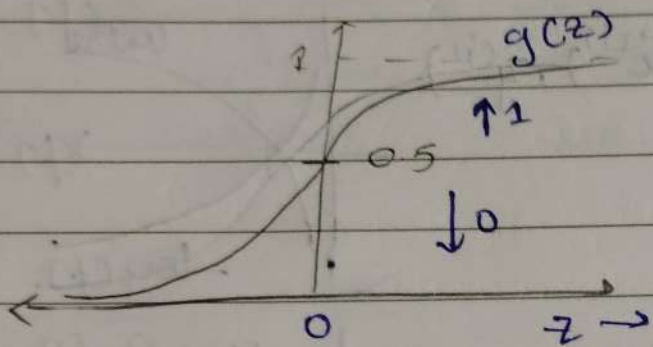
Polynomial Regression

more than two parameters

Classification

Binary Classification — Yes or No
 true false

1 0



Sigmoid function
 $g(z) = \frac{1}{1 + e^{-z}}$

$$0 < g(z) < 1$$

Logistic Regression (Classification).

$$z = \vec{w} \cdot \vec{x} + b$$

$$g(z) = \frac{1}{1 + e^{-z}} = \frac{1}{1 + e^{-(\vec{w} \cdot \vec{x} + b)}}$$

$$f(x) = g(\vec{w} \cdot \vec{x} + b)$$

$$\text{if } \vec{w} \cdot \vec{x} + b > 0$$

$$\hat{y} = 1$$

$$\vec{w} \cdot \vec{x} + b < 0$$

$$\hat{y} = 0$$

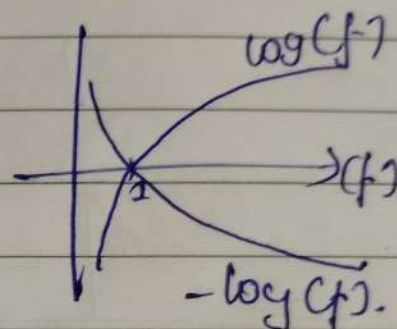
For two parameters. $g = w_1 x_1 + w_2 x_2 + b$

Logistic Loss Function

$$L(f_{w,b}(x^{(i)}), y^{(i)}) = \begin{cases} -\log(f_{w,b}(x^{(i)})) & \text{if } y^{(i)} = 0 \\ -\log(1 - f_{w,b}(x^{(i)})) & \text{if } y^{(i)} = 1 \end{cases}$$

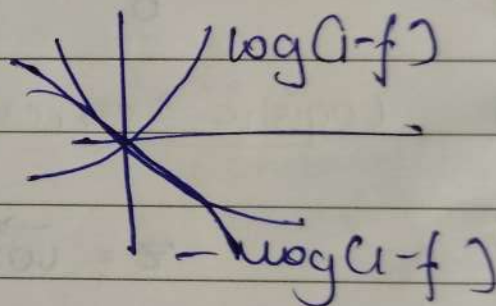
cost function

$$J(w, b) = \frac{1}{m} \sum_{i=1}^m L(f_{w,b}(\vec{x}^{(i)}), y^{(i)})$$



Simplified Loss Function

$$L(f_{w,b}(x^{(i)}), y^{(i)}) = -y^{(i)} \log(f_{w,b}(x^{(i)})) - (1 - y^{(i)}) \log(1 - f_{w,b}(x^{(i)}))$$



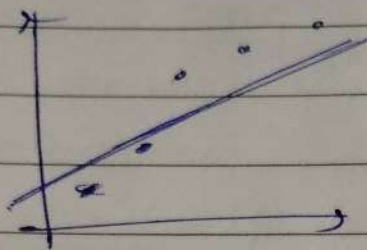
$$J(w, b) = \frac{1}{m} \sum_{i=1}^m [L(f(x^{(i)}), y^{(i)})]$$

(average of loss function)

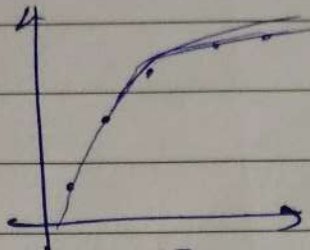
$$J(w, b) = -\frac{1}{m} \sum_{i=1}^m [y^{(i)} \log(f) + (1 - y^{(i)}) \log(1 - f)]$$

Linear Regression - $f_{w,b}(\vec{x}) = \vec{w} \cdot \vec{x} + b$

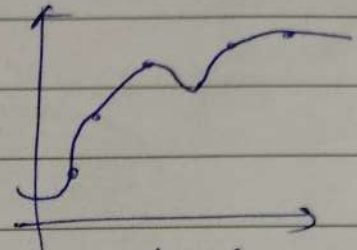
Logistic Regression - $f_{w,b}(\vec{x}) = \frac{1}{1 + e^{-(\vec{w} \cdot \vec{x} + b)}}$

Overfitting

underfitting
High Bias



Generalization
Just Right



overfitting
High Variance

Regularisation

Reducing the value of parameters w_j .

$$\min_{w, b} J(w, b) = \underbrace{\frac{1}{2m} \sum_{i=1}^m (f(w, b)(x^{(i)}) - y^{(i)})^2}_{\text{mean squared error} \rightarrow \text{fit data}} + \underbrace{\frac{\lambda}{2m} \sum_{j=1}^m w_j^2}_{\text{regularisation term} \rightarrow \text{keep } w_j \text{ small}}$$

λ balances both the goals

—END—